DISCRETE MATH · THE LOGIC OF COMPOUND STATEMENTS

Truth Table Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Truth Table Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. How many rows does a truth table need for a statement with 3 variables? With 4?

2. Predict: is "if p then q" equivalent to "if q then p"? Explain in a sentence.

2 Explore use the tool

1. Statement 1: p → q. Use the buttons to make the converse, inverse and contrapositive. Which are equivalent to p → q?

2. Compare the converse and the inverse with each other (type q → p and ~p → ~q). What do you find?

3. Type ~(p → q) and find an equivalent statement with no arrow. Record it.

4. Test an argument mode: p ∨ q; ~p, so q. How many critical rows are there, and is it valid?

3 Record use the tool and fill in the table

Classify each statement

Use Statement 1 only. Write tautology, contradiction or contingency, and the number of true rows.

Statement	Rows	True rows	Type
p ∨ ~p			
p ∧ ~p			
(p → q) ∨ (q → p)			
p ↔ q			
(p ∧ q) → r			

Valid or invalid?

Premises	Conclusion	Critical rows	Valid?
p → q; p	q		
p → q; q	p		
p → q; ~p	~q		
p → q; q → r	p → r		

4 Explain what did you discover?

1. Two statements are logically equivalent when their columns agree in _____ row.
2. p → q is equivalent to its _____, but not to its converse or its _____.
3. ~(p → q) is equivalent to _____.
4. An argument form is valid when the conclusion is _____ in every critical row (every row where all the premises are _____).
5. A classmate says, "My argument has a true conclusion in one critical row, so it is valid." Explain the mistake.

5 Apply a new situation, no tool

1. Write the contrapositive of "If a number is divisible by 6, then it is even." Is it true?
2. Write the negation of "If it rains, then the game is canceled" without using "if".
3. Is (p → q) ∧ (q → r) → (p → r) a tautology? How many rows do you need to check?
4. "If I study, I pass. I passed. So I studied." Name the form and say whether it is valid.